

Segundo Taller

martes, 25 de noviembre de 2025 8:12 p. m.

1. Realice un muestreo aleatorio estratificado sin reemplazo sobre la información en análisis en función de la cobertura del ingreso. Emplee la **afijación proporcional y afijación de Neymann**. ¿Las unidades muestrales intra-estrato son homogéneas? ¿Los estratos son suficientemente heterogéneos entre sí?

Parámetro a estimar	Estimador puntual	Varianza estimada del estimador	Intervalo de confianza del (1 - α) 100%
μ	$\bar{y}_{st} = \frac{1}{N} \sum_{i=1}^L N_i \bar{y}_i$	$\hat{V}(\bar{y}_{st}) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \hat{V}(\bar{y}_i)$	$\bar{y}_{st} \pm t_{\alpha/2, n-L} \sqrt{\hat{V}(\bar{y}_{st})}$
τ	$\hat{\tau}_{st} = N \bar{y}_{st} = \sum_{i=1}^L \hat{\tau}_i$	$\hat{V}(\hat{\tau}_{st}) = N^2 \hat{V}(\bar{y}_{st})$	$N \bar{y}_{st} \pm t_{\alpha/2, n-L} N \sqrt{\hat{V}(\bar{y}_{st})}$
p	$\hat{p}_{st} = \frac{1}{N} \sum_{i=1}^L N_i \hat{p}_i$	$\hat{V}(\hat{p}_{st}) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \hat{V}(\hat{p}_i)$	$\hat{p}_{st} \pm t_{\alpha/2, n-L} \sqrt{\hat{V}(\hat{p}_{st})}$
A	$\hat{A}_{st} = N \hat{p}_{st} = \sum_{i=1}^L \hat{A}_i$	$\hat{V}(\hat{A}_{st}) = N^2 \hat{V}(\hat{p}_{st})$	$N \hat{p}_{st} \pm t_{\alpha/2, n-L} N \sqrt{\hat{V}(\hat{p}_{st})}$

Donde: $\bar{y}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij}$, $\hat{p}_i = \frac{a_i}{n_i}$, $S_i^2 = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2$ y $n - L = \sum_{i=1}^L (n_i - 1)$

$\hat{V}(\bar{y}_i) = \left(\frac{N_i - n_i}{N_i} \right) \frac{S_i^2}{n_i}$, $\hat{V}(\hat{p}_i) = \left(\frac{N_i - n_i}{N_i} \right) \frac{\hat{p}_i (1 - \hat{p}_i)}{n_i - 1}$

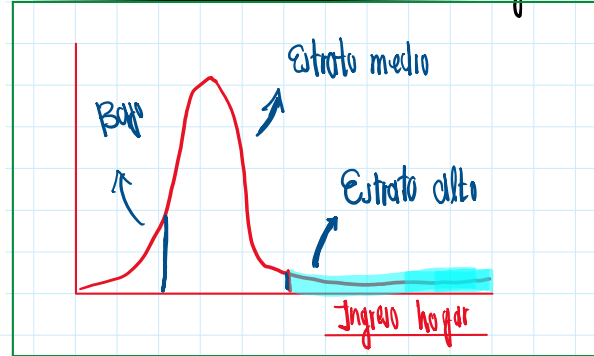
$$n_i = \frac{N_i \sigma_i}{\sum_{k=1}^L N_k \sigma_k} \cdot n$$

Neyman

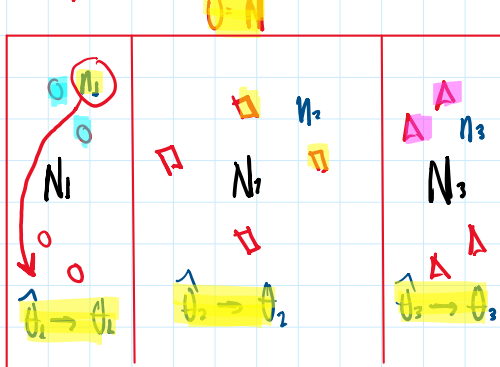
$$n_i = \left(\frac{N_i}{N} \right) \cdot n$$

proporcional

- Las unidades intra-estrato no son homogéneas



Proporcional: N_i : tamaño de la i -ésima población



$\sum_{i=1}^L N_i = N$; $1 = \text{número de estratos}$

$\hat{\theta}_{st} = \sum_{i=1}^L \hat{\theta}_i \cdot m_i$; $m_i = \frac{N_i}{N}$

$\sum_{i=1}^L n_i = n$ $\rightarrow$ n global

$n_i = \frac{N_i}{N} \cdot n$

Neyman:

$\hat{y}_i = \frac{N_i \cdot \sigma_i}{\sum_{k=1}^L N_k \cdot \sigma_k}$

- (1). Existe heterogeneidad entre estratos
(2). Existe homogeneidad dentro de los estratos

Neymann: $N_1 = 39294$; $N_2 = 39157$; $N_3 = 3510$

`> N_i`
[1] 39294 39157 3510

$N = \sum_{i=1}^L N_i = 81961$

$n = 8000$

Proporcional: $n_i = \left(\frac{N_i}{N} \right) \cdot n = \left(\frac{39294}{81961} \right) \cdot 8000$

`> n_i`
[1] 3835 3885 3822 813 342 602

`> n_i`
[1] 3835 3822 342

$\sum_{i=1}^L n_i = n = 8000$

Neymann:

σ_i ; $\sigma^2 = \frac{y_{\max} - y_{\min}}{6}$

Estimación Varianza

y_i : El ingreso del i -ésimo hogar

`> sigma_i`
[1] 2035220 3677480 9020582

$n_1 = \frac{N_1 \cdot \sigma_1}{\sum_{k=1}^L N_k \cdot \sigma_k} \cdot n = \frac{N_1 \cdot \sigma_1}{(N_1 \cdot \sigma_1 + N_2 \cdot \sigma_2 + N_3 \cdot \sigma_3)} \cdot n$

$n_1 = 39294 \cdot (2035220)$

$[39294 \cdot (2035220) + 39157 \cdot (3677480) + \dots]$

`> n_iNeymann`
[1] 2502.7077 4506.4277 990.8646

$\sum_{i=1}^L n_i = n$

2. Estime el ingreso total de los hogares τ , así como el ingreso promedio percibido μ . Calcule los correspondientes intervalos de confianza. Interprete estos resultados.

$\hat{\tau}_{st}$; $\hat{\tau}_{st} \rightarrow \tau_{st}$; $\hat{\theta}_{st} \rightarrow \theta_{st}$; $\hat{\theta}_{st} = \sum_{i=1}^L \hat{\theta}_i \cdot m_i$; **IC:** $\hat{\theta}_{st} \pm t_{\alpha/2, n-L} \cdot \sqrt{\text{Var}(\hat{\theta}_{st})}$

Calcule los correspondientes intervalos de confianza. Interprete estos resultados.

• $\hat{\tau}_{st}$; $\hat{\tau}_{st} \rightarrow \tau_{st}$; $\hat{\theta}_{st} \rightarrow \theta_{st}$; $\hat{\theta}_{st} = \sum_{i=1}^L \hat{\theta}_i \cdot m_i$; IC: $\hat{\theta}_{st} \pm t_{\alpha/2, n-L} \cdot \sqrt{\widehat{Var}(\hat{\theta}_{st})}$

• $\widehat{Var}(\hat{\theta}_{st}) = \widehat{Var}(\sum_{i=1}^L \hat{\theta}_i \cdot m_i) = \widehat{Var}(\sum_{i=1}^L \hat{\theta}_i \cdot \frac{N_i}{N}) = \sum_{i=1}^L \frac{N_i^2}{N^2} \widehat{Var}(\hat{\theta}_i)$ (MAS)

Parámetro a estimar	Varianza estimada del estimador	Intervalo de confianza del $(1 - \alpha)100\%$
μ	$\hat{V}(\bar{Y}) = \left(\frac{N-n}{N}\right) \frac{S^2}{n}$	$\bar{Y} \pm t_{\alpha/2, n-1} \sqrt{\hat{V}(\bar{Y})}$
τ	$\hat{V}(\hat{\tau}) = N^2 \hat{V}(\bar{Y})$	$N\bar{Y} \pm t_{\alpha/2, n-1} N \sqrt{\hat{V}(\bar{Y})}$
ρ	$\hat{V}(\hat{\rho}) = \left(\frac{N-n}{N}\right) \frac{\hat{\rho}(1-\hat{\rho})}{n-1}$	$\hat{\rho} \pm t_{\alpha/2, n-1} \sqrt{\hat{V}(\hat{\rho})}$
A	$\hat{V}(\hat{A}) = N^2 \hat{V}(\hat{\rho})$	$N\hat{\rho} \pm t_{\alpha/2, n-1} N \sqrt{\hat{V}(\hat{\rho})}$

Donde $S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$ es la varianza muestral de la variable cuantitativa Y , y $t_{n-1, \alpha/2}$ es el cuantil $1 - \alpha/2$ de la distribución t-Student con $n - 1$ grados de libertad.

• No vamos a considerar la varianza sino la varianza estimada.
 $\widehat{Var}(\hat{\theta}_{st}) = \sum_{i=1}^L \frac{N_i^2}{N^2} \cdot \widehat{Var}(\hat{\theta}_i)$

$\sum_{i=1}^L \hat{\theta}_i \cdot m_i \pm t_{\alpha/2, n-L} \cdot \sqrt{\sum_{i=1}^L \frac{N_i^2}{N^2} \cdot \widehat{Var}(\hat{\theta}_i)}$

$\hat{\theta}_{st} = 100$; IC₉₅ = 100,3

• $\hat{\mu}_{st}$; $\hat{\mu}_{st} = \bar{Y}_{st} \rightarrow \mu_{st}$; $\bar{Y}_{st} = \sum_{i=1}^L \bar{Y}_i \cdot m_i$
 $\bar{Y}_i = \text{Promedio estimado de la primera población}$
 $\bar{Y}_{st} = \sum_{i=1}^L \bar{Y}_i \cdot \frac{N_i}{N}$

$\bar{Y}_1 = \sum_{i=1}^{n_1} \frac{Y_{i1}}{n_1}$; $\bar{Y}_2 = \sum_{i=1}^{n_2} \frac{Y_{i2}}{n_2}$; $\bar{Y}_3 = \sum_{i=1}^{n_3} \frac{Y_{i3}}{n_3}$

$\bar{Y}_{st} = \bar{Y}_1 \left(\frac{N_1}{N}\right) + \bar{Y}_2 \left(\frac{N_2}{N}\right) + \bar{Y}_3 \left(\frac{N_3}{N}\right)$

> n.1			
[1]	3836	3822	342
	n_1	n_2	n_3

$\bar{Y}_1 =$	\$Estimador [1] 1315553
$\bar{Y}_2 =$	\$Estimador [1] 2449254
$\bar{Y}_3 =$	\$Estimador [1] 7408515

(Baja) E_1

(Medio) E_2

(Alto) E_3

> obj_promedio
\$Estimador [1] 2118113
\$Varianza_Estimador [1] 1062253431
\$IC [1] 2054224 2182002

IC: $\bar{Y}_{st} \pm t_{\alpha/2, n-L} \cdot \sqrt{\sum_{i=1}^L \frac{N_i^2}{N^2} \cdot \left(\frac{N_i - n_i}{N_i}\right) \frac{S_i^2}{n_i}}$

Parámetro a estimar	Estimador puntual	Varianza estimada del estimador	Intervalo de confianza del $(1 - \alpha)100\%$
μ	$\bar{Y}_{st} = \frac{1}{N} \sum_{i=1}^L N_i \bar{Y}_i$	$\hat{V}(\bar{Y}_{st}) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \hat{V}(\bar{Y}_i)$	$\bar{Y}_{st} \pm t_{\alpha/2, n-L} \sqrt{\hat{V}(\bar{Y}_{st})}$
τ	$\hat{\tau}_{st} = N \bar{Y}_{st} = \sum_{i=1}^L \tau_i$	$\hat{V}(\hat{\tau}_{st}) = N^2 \hat{V}(\bar{Y}_{st})$	$N\bar{Y}_{st} \pm t_{\alpha/2, n-L} N \sqrt{\hat{V}(\bar{Y}_{st})}$
ρ	$\hat{\rho}_{st} = \frac{1}{N} \sum_{i=1}^L N_i \hat{\rho}_i$	$\hat{V}(\hat{\rho}_{st}) = \frac{1}{N^2} \sum_{i=1}^L N_i^2 \hat{V}(\hat{\rho}_i)$	$\hat{\rho}_{st} \pm t_{\alpha/2, n-L} \sqrt{\hat{V}(\hat{\rho}_{st})}$
A	$\hat{A}_{st} = N \hat{\rho}_{st} = \sum_{i=1}^L \hat{A}_i$	$\hat{V}(\hat{A}_{st}) = N^2 \hat{V}(\hat{\rho}_{st})$	$N\hat{\rho}_{st} \pm t_{\alpha/2, n-L} N \sqrt{\hat{V}(\hat{\rho}_{st})}$

Donde: $\bar{Y}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} Y_{ij}$; $\hat{\rho}_i = \frac{a_i}{n_i}$; $S_i^2 = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_i)^2$ y $n - L = \sum_{i=1}^L (n_i - 1)$
 $\hat{V}(\bar{Y}_i) = \left(\frac{N_i - n_i}{N_i}\right) \frac{S_i^2}{n_i}$; $\hat{V}(\hat{\rho}_i) = \left(\frac{N_i - n_i}{N_i}\right) \frac{\hat{\rho}_i (1 - \hat{\rho}_i)}{n_i - 1}$

• $\hat{\tau}_{st} \rightarrow \tau_{st}$: $\hat{\tau}_{st} = \sum_{i=1}^L \hat{\tau}_i \cdot m_i$
 $\hat{\tau}_i = N_i \bar{Y}_i$; $\hat{\tau}_1 = N_1 \bar{Y}_1$; $\hat{\tau}_2 = N_2 \bar{Y}_2$
 $\hat{\tau}_{st} = N \bar{Y}_{st}$

$\hat{\tau}_{st} = 81961 \cdot 2118113$

> obj_total
\$Estimador [1] 173602676428
\$Varianza_Estimador [1] 7.1358e+18
\$IC [1] 168366247703 178839105153

• $\hat{\tau}_{st} \pm t_{\alpha/2, n-L} \cdot \sqrt{\widehat{Var}(\hat{\tau}_{st})}$
 $N \cdot \bar{Y}_{st} \pm t_{\alpha/2, n-L} \cdot \sqrt{\widehat{Var}(\hat{\tau}_{st})}$
 $\widehat{Var}(\hat{\tau}_{st}) = \widehat{Var}(N \cdot \bar{Y}_{st}) = N^2 \cdot \widehat{Var}(\bar{Y}_{st})$

$N \cdot \bar{y}_{st} \pm t_{\alpha/2, n-1} \cdot N \sqrt{\sum_{i=1}^L \frac{N_i^2}{N^2} \cdot \left(\frac{N_i - n_i}{N_i}\right) \cdot \frac{S_i^2}{n_i}}$
 $N \cdot IC_{\bar{y}_{st}} = 81461 \cdot [IC_{\bar{y}_{st}}]$

$Var[\hat{y}_{st}] = Var[N \cdot \bar{y}_{st}] = N^2 \cdot Var[\bar{y}_{st}]$
 $Var[\bar{y}_{st}] = \sum_{i=1}^L \frac{N_i^2}{N^2} \cdot Var[\bar{y}_i]$
 $= \sum_{i=1}^L \frac{N_i^2}{N^2} \cdot \left(\frac{N_i - n_i}{N_i}\right) \cdot \frac{S_i^2}{n_i}$

3. Estime el total de hogares que poseen automovil A, así como la proporción de hogares que se perciben como seguros p. Calcule los correspondientes intervalos de confianza. Interprete estos resultados.

$I_{un}(i) = \begin{cases} 1, & i \in A \\ 0, & i \notin A \end{cases}$

$Y_i \sim Ber(p); Y_i = \begin{cases} 1 \\ 0 \end{cases}; \hat{A}_{st} \rightarrow A_{st}; \hat{A}_{st} = \sum_{i=1}^L \hat{A}_i \cdot m_i$

Construir Intervalos confianza: $\hat{p}_{st} \rightarrow p_{st}; \hat{p}_{st} = \sum_{i=1}^L \hat{p}_i \cdot m_i$

$\hat{A}_{st} \pm t_{\alpha/2, n-1} \cdot \sqrt{Var(\hat{A}_{st})}$
 $\hat{p}_{st} \pm t_{\alpha/2, n-1} \cdot \sqrt{Var(\hat{p}_{st})}$

$\hat{A}_{st} = N \cdot \hat{p}_{st}; IC_{\hat{A}_{st}} = N IC_{\hat{p}_{st}} \checkmark$

Construir estimadores: $\hat{A}_{st} = \hat{A}_1 \cdot m_1 + \hat{A}_2 \cdot m_2 + \hat{A}_3 \cdot m_3$

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> objeto_A
$Estimador
[1] 6815.659

$Varianza_Estimador
[1] 51822.41

$IC
[1] 6369.414 7261.903
    
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$\hat{p}_{st} = \hat{p}_1 \cdot m_1 + \hat{p}_2 \cdot m_2 + \hat{p}_3 \cdot m_3$

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> objeto_SEC
$Estimador
[1] 0.8472565

$Varianza_Estimador
[1] 1.455595e-05

$IC
[1] 0.8397776 0.8547353
    
```

4. Determine el tamaño de muestra mínimo necesario para estimar la proporción de personas que poseen propiedad recreativa con un límite para el error de estimación del 2% y una confianza del 95%. Para los p_i utilice las estimaciones intra-estrato y para el valor de la afijación ψ_i emplee la afijación proporcional.

$n = \sum_{i=1}^L \frac{N_i \sigma_i^2}{V_i}$
 $N^2 D + \sum_{i=1}^L N_i \sigma_i^2$

$V_i = \left(\frac{N_i}{N}\right)$

$i=1,2,3; L=3$

$\sigma^2 = \frac{y_{max} - y_{min}}{6}$

$D = \frac{\delta^2}{Z_{\alpha/2}}$

$\sigma_i^2 = p_i(1-p_i)$

```

> p_i
[1] 0.05891554 0.09759288 0.25146199
    
```

p_1, p_2, p_3

$n = \sum_{i=1}^L \frac{N_i p_i(1-p_i)}{N^2 \left(\frac{\delta^2}{Z_{\alpha/2}^2}\right) + \sum_{i=1}^L N_i p_i(1-p_i)}$

$\hat{p}_1, \hat{p}_2, \hat{p}_3$

$D = \frac{(0.02)^2}{(1.96)^2}$

$\alpha=0.05; Z_{\alpha/2}=1.96$
 $n = 730.2009$

$n = 730.2009$

$\frac{(0.06)^2}{(1.96)^2}$

$n_1, n_2, n_3; \sum_{i=1}^3 n_i = n$

$n_i = \sqrt{f_i} \cdot n$

```
> psi_i[1] * n
[1] 350.0752 n1
> psi_i[2] * n
[1] 348.8547 n2
> psi_i[3] * n
[1] 31.27103 n3
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$n_1 = 351, n_2 = 349, n_3 = 32$

$n_{red,nd} = 732$

(Nuevo n)